M3 SIGMOID STEP 2: COLUMN S-CURVE & BASELINE OPTIMIZATION

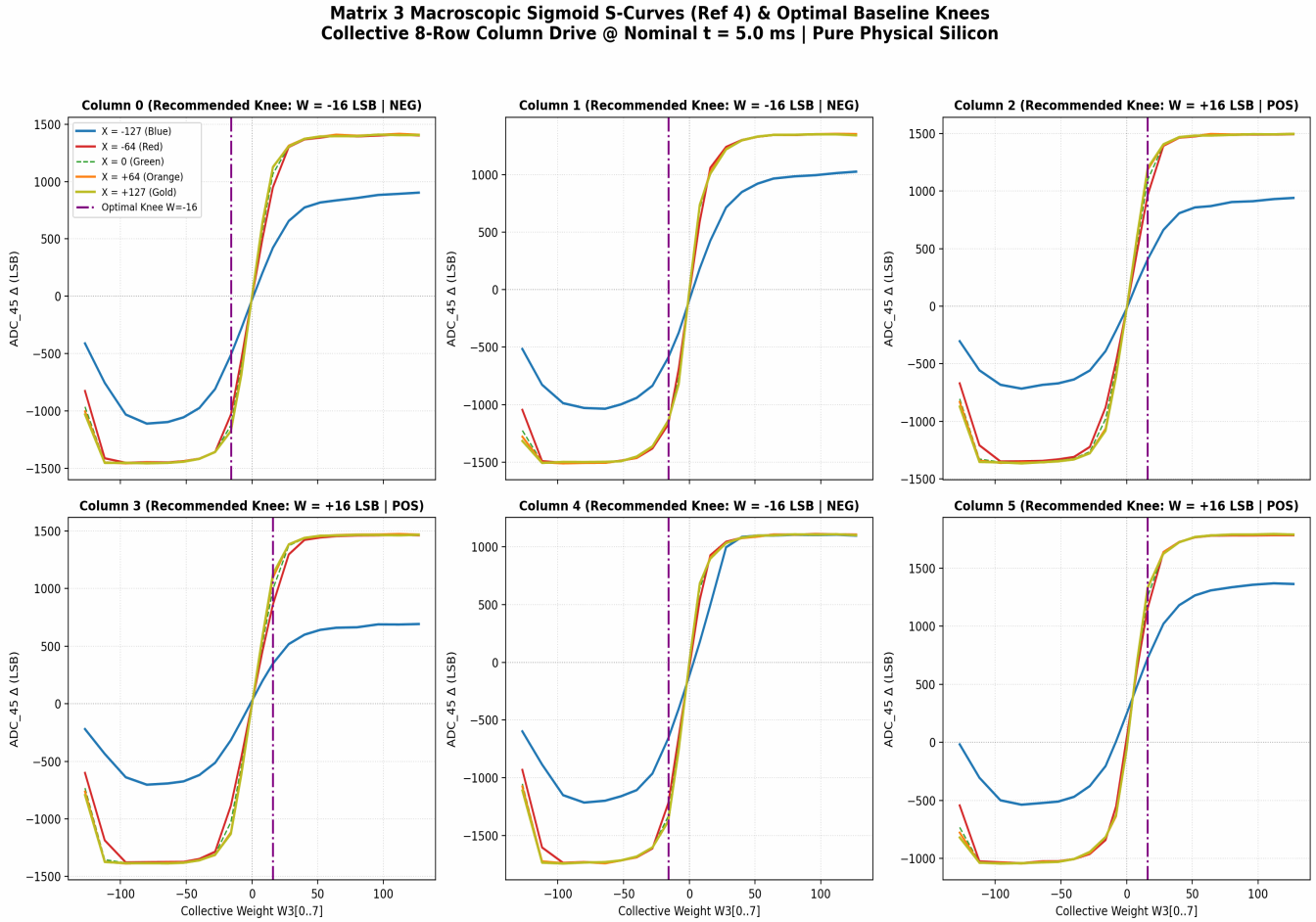
Macroscopic Sigmoid S-Curve Characterization (Ref 4) on Differential ADC_45

Optimal Baseline Initial Weight Vector $W_{3,baseline}$ for Inverted MAML

EXECUTIVE SUMMARY & OPERATIONAL FINDINGS:

Steg 2 drives all rows on each column collectively (W3[0..7][c] ramped simultaneously) through the ColTIA and analog sigmoid activation network (Ref 4). By testing arbitrary medium-scale signals (X = ±64) alongside full-scale signals (X = ±127), we observe the true macroscopic S-curve and determine the precise baseline weight bias point W_base[c] where sensitivity is maximal and stochastic jitter is minimal. This calibrated vector will serve as the starting weights for compensation matrices and the center for future L2 regularization.

1. Macroscopic Column S-Curves (X = ±127, ±64)



2. Calibrated Baseline Weight Vector $W_{3,baseline}$

Column	Optimal Knee W_{base}	Best Polarity	Sensitivity @ X=±64	Dynamic Swing (LSB)	Regularization Anchor
C0	**-16 LSB**	**NEG (-)**	47.35 LSB/W	2861 LSB	Pull toward -16
C1	**-16 LSB**	**NEG (-)**	43.45 LSB/W	2859 LSB	Pull toward -16
C2	**+16 LSB**	**POS (+)**	51.31 LSB/W	2856 LSB	Pull toward +16
C3	**+16 LSB**	**POS (+)**	48.99 LSB/W	2861 LSB	Pull toward +16
C4	**-16 LSB**	**NEG (-)**	56.47 LSB/W	2847 LSB	Pull toward -16

C5	**+16 LSB**	**POS (+)**	57.75 LSB/W	2833 LSB	Pull toward +16

3. Recommended Global Baseline Configuration

Calibrated Vector: [C0: -16, C1: -16, C2: +16, C3: +16, C4: -16, C5: +16]
By setting the compensation weights at these initial knee biases instead of 0, the analog gradient feedback immediately enters the responsive non-linear regime with maximum dynamic sensitivity and zero dead-band delay.